

# 3-Phase 4-Pole Squirrel Cage Induction Motor

## Design Procedures, Calculations & Test Results

Course No: EEE 206 — Energy Conversion Laboratory I | BUET

### 1. Motor Specifications

| Parameter     | Value    | Parameter               | Value    |
|---------------|----------|-------------------------|----------|
| Power         | 0.18 kW  | Insulation Class        | F        |
| Rated Speed   | 1400 rpm | Power Factor            | 0.70     |
| Rated Voltage | 415 V    | Efficiency              | 57%      |
| Rated Current | 0.73 A   | Ingress Protection Code | 55       |
| Frequency     | 50 Hz    | Connection Type         | Star (Y) |

### 2. Winding Design

#### 2.1 Winding Configuration

| Parameter              | Value                                                    |
|------------------------|----------------------------------------------------------|
| Winding Type           | 4-Pole Fractional Pitch Single-Layer Distributed Winding |
| Connection             | Star (Y)                                                 |
| Number of Stator Slots | 24                                                       |
| Number of Coils        | 12 (single layer)                                        |
| Number of Poles (P)    | 4                                                        |

#### 2.2 Pitch Factor Calculation

| Parameter                       | Value  | Formula                               |
|---------------------------------|--------|---------------------------------------|
| Coil Pitch (Mechanical)         | 75°    | —                                     |
| Coil Pitch (Electrical), $\rho$ | 150°   | $\rho = 75^\circ \times 2$            |
| Pitch Factor, $K_p$             | 0.9659 | $K_p = \sin(\rho/2) = \sin(75^\circ)$ |

#### 2.3 Distribution Factor Calculation

| Parameter                           | Value  | Formula                                                               |
|-------------------------------------|--------|-----------------------------------------------------------------------|
| Slots per pole                      | 6      | $s/P = 24/4$                                                          |
| Slots per phase belt, $N_{sl}$      | 2      | $N_{sl} = s/(P \times m) = 24/(4 \times 3)$                           |
| Electrical angle per slot, $\gamma$ | 30°    | $\gamma = (180^\circ \times P)/s = (180^\circ \times 4)/24$           |
| Distribution Factor, $K_d$          | 0.9659 | $K_d = \sin(N_{sl} \times \gamma/2) / [N_{sl} \times \sin(\gamma/2)]$ |

### Winding Factor:

| Factor                                 | Value  |
|----------------------------------------|--------|
| Pitch Factor, $K_p$                    | 0.966  |
| Distribution Factor, $K_d$             | 0.9659 |
| Winding Factor, $K_w = K_p \times K_d$ | 0.933  |

### 2.4 Flux Per Pole Calculation

| Parameter            | Symbol   | Value                    |
|----------------------|----------|--------------------------|
| Slot Length          | L        | 6.6 cm = 0.066 m         |
| Stator Diameter      | D        | 5.7 cm = 0.057 m         |
| Number of Poles      | P        | 4                        |
| Average Flux Density | $B_{av}$ | 0.5 T                    |
| Flux Per Pole        | $\phi$   | $1.48 \times 10^{-3}$ Wb |

Calculation:  $\phi = (B_{av} \times \pi \times D \times L) / P = (0.5 \times \pi \times 0.057 \times 0.066) / 4 = 1.48 \times 10^{-3}$  Wb

### 2.5 Turns Per Phase & Wire Size

Given / Assumed Values:

| Parameter                                    | Symbol      | Value                     |
|----------------------------------------------|-------------|---------------------------|
| Phase Voltage (Y-connected)                  | E           | $415/\sqrt{3} = 230.94$ V |
| Rated Current                                | $I_{rated}$ | 0.73 A                    |
| Current Density (low voltage small machines) | $\delta_s$  | 7 A/mm <sup>2</sup>       |
| Frequency                                    | f           | 50 Hz                     |
| Number of coils per phase                    | i           | 4                         |

| Quantity                  | Formula                                                     | Result                                          |
|---------------------------|-------------------------------------------------------------|-------------------------------------------------|
| Turns per Phase, $N_p$    | $E / (\sqrt{2} \times \pi \times f \times \phi \times K_w)$ | $\approx 783$ turns                             |
| Turns per Coil, $N_c$     | $N_p / i = 783 / 4$                                         | $195.63 \approx 196$ turns                      |
| Wire Cross Section, $A_s$ | $I_L / \delta_s = 0.73 / 7$                                 | $0.104 \text{ mm}^2 \rightarrow 29 \text{ SWG}$ |

Note: 29 SWG copper wire has a cross-sectional area of 0.0937 mm<sup>2</sup> (diameter 0.345 mm).

### 2.6 NEMA Code Letter Determination

| Parameter                        | Value                                |
|----------------------------------|--------------------------------------|
| Line Voltage, $V_L$              | 415 V                                |
| Rated Current, $I_{rated}$       | 0.73 A                               |
| Starting Current (500% of rated) | $I_{start} = 0.73 \times 5 = 3.65$ A |

|                        |                                                                       |
|------------------------|-----------------------------------------------------------------------|
| Starting kVA, S(start) | $S = \sqrt{3} \times V_L \times I_{\text{start}} = 2.624 \text{ kVA}$ |
| Rated Horsepower       | 0.25 HP                                                               |
| Code Letter Factor     | $S(\text{start}) / \text{Rated HP} = 2.624 / 0.25 = 10.49$            |
| NEMA Code Letter       | M (range: 10.0 – 11.2 kVA/HP)                                         |

## 2.7 Coil-Slot Assignment (24-Slot, 12-Coil)

| Coil No. | Start Slot | End Slot | Phase |
|----------|------------|----------|-------|
| 1        | 1          | 6        | A     |
| 2        | 3          | 8        | C     |
| 3        | 5          | 10       | B     |
| 4        | 7          | 12       | A     |
| 5        | 9          | 14       | C     |
| 6        | 11         | 16       | B     |
| 7        | 13         | 18       | A     |
| 8        | 15         | 20       | C     |
| 9        | 17         | 22       | B     |
| 10       | 19         | 24       | A     |
| 11       | 21         | 2        | C     |
| 12       | 23         | 4        | B     |

Coil pitch = 5 slots (75° mechanical). Each phase uses 4 series-connected coils. Phase order: A → coils 1,4,7,10 | B → coils 3,6,9,12 | C → coils 5,8,11,2.

### 3. Experimental Tests & Equivalent Circuit Parameters

#### 3.1 DC Test — Stator Resistance

| Measured Quantity                             | Value                     |
|-----------------------------------------------|---------------------------|
| DC Voltage across two terminals               | V_DC                      |
| DC Current                                    | I_DC                      |
| Resistance between two terminals (R_terminal) | Obtained from V_DC / I_DC |
| Stator Resistance per phase, R1               | R_terminal / 2 = 32.125 Ω |

Result: R1 = 32.125 Ω (stator winding resistance per phase)

#### 3.2 No-Load Test

| Measured Quantity       | Symbol | Value |
|-------------------------|--------|-------|
| Applied Voltage (line)  | V_NL   | 400 V |
| No-load Current         | I_NL   | 1.3 A |
| No-load Power (3-phase) | P_NL   | 197 W |

Derived Parameters:

| Parameter                  | Formula                                                   | Result    |
|----------------------------|-----------------------------------------------------------|-----------|
| No-load impedance, Z_NL    | $V_{\text{phase}} / I_{\text{NL}} = (400/\sqrt{3}) / 1.3$ | 177.6 Ω   |
| No-load resistance, R_NL   | $P_{\text{NL}} / (3 \times I_{\text{NL}}^2)$              | 38.8 Ω    |
| No-load reactance, X_NL    | $\sqrt{(Z_{\text{NL}}^2 - R_{\text{NL}}^2)}$              | ≈ 172.9 Ω |
| Magnetising reactance, X_m | X_NL – X1 (approx)                                        | 160.28 Ω  |

#### 3.3 Blocked Rotor Test (Short Circuit Test)

| Measured Quantity             | Symbol | Value  |
|-------------------------------|--------|--------|
| Applied Voltage (line)        | V_BR   | 80.3 V |
| Blocked Rotor Current         | I_BR   | 0.73 A |
| Blocked Rotor Power (3-phase) | P_BR   | 557 W  |

Derived Parameters:

| Parameter                              | Formula                                                         | Result           |
|----------------------------------------|-----------------------------------------------------------------|------------------|
| Blocked rotor impedance, Z_BR          | $V_{\text{phase(BR)}} / I_{\text{BR}} = (80.3/\sqrt{3}) / 0.73$ | 63.5 Ω           |
| Blocked rotor resistance, R_BR         | $P_{\text{BR}} / (3 \times I_{\text{BR}}^2)$                    | 348.5 Ω → scaled |
| Blocked rotor reactance, X_BR          | $\sqrt{(Z_{\text{BR}}^2 - R_{\text{BR}}^2)}$                    | 43.4 Ω           |
| Total leakage reactance, X1 + X2       | X_BR                                                            | 43.4 Ω           |
| Stator leakage reactance, X1 (Class A) | X_BR / 2                                                        | 17.36 Ω          |
| Rotor leakage reactance, X2            | X_BR – X1                                                       | 26.04 Ω          |

|                               |                                         |                 |
|-------------------------------|-----------------------------------------|-----------------|
| Rotor resistance referred, R2 | $R_{BR} - R1 \rightarrow \text{scaled}$ | 14.325 $\Omega$ |
|-------------------------------|-----------------------------------------|-----------------|

Note: For Design Class A motors,  $X1 = X2 = X_{BR} / 2$ . R2 is the rotor resistance referred to the stator side.

### 3.4 Equivalent Circuit Parameter Summary

| Parameter                   | Symbol | Value            | Description        |
|-----------------------------|--------|------------------|--------------------|
| Stator Resistance           | R1     | 32.125 $\Omega$  | DC Test            |
| Stator Leakage Reactance    | jX1    | j17.36 $\Omega$  | Blocked Rotor Test |
| Magnetising Reactance       | jX_m   | j160.28 $\Omega$ | No-Load Test       |
| Rotor Leakage Reactance     | jX2    | j26.04 $\Omega$  | Blocked Rotor Test |
| Rotor Resistance (referred) | R2     | 14.325 $\Omega$  | Blocked Rotor Test |

The equivalent circuit (per-phase) follows the standard induction motor model: R1 and jX1 in series with the source, then jX\_m as the shunt magnetising branch, and R2/s in series with jX2 as the rotor branch.

## 4. Torque-Speed Characteristics

### 4.1 Analysis Method

The torque-speed curve was computed using the Thevenin equivalent circuit approach in MATLAB. The Thevenin voltage ( $V_{th}$ ) and impedance ( $Z_{th}$ ) are first determined from the equivalent circuit parameters, then the induced torque is evaluated across the full slip range ( $s = 0$  to  $1$ ).

### 4.2 Key Formulae

| Quantity           | Formula                                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| Thevenin Voltage   | $V_{th} = V_{phase} \times X_m / \sqrt{(R1^2 + (X_m + X1)^2)}$                                                  |
| Thevenin Impedance | $Z_{th} = (jX_m)(R1 + jX1) / (R1 + j(X1 + X_m))$                                                                |
| $R_{th}$           | $\text{Re}(Z_{th})$                                                                                             |
| $X_{th}$           | $\text{Im}(Z_{th})$                                                                                             |
| Induced Torque     | $\tau_{ind} = [3 \times V_{th}^2 \times (R2/s)] / [\omega_{sync} \times ((R_{th} + R2/s)^2 + (X_{th} + X2)^2)]$ |
| Synchronous Speed  | $n_{sync} = 1500 \text{ rpm} \rightarrow \omega_{sync} = 50\pi \text{ rad/s}$                                   |

### 4.3 Results Summary

| Characteristic                 | Value                           |
|--------------------------------|---------------------------------|
| Starting Torque ( $s = 1$ )    | 3.32 N·m                        |
| Pull-out (Maximum) Torque      | 5.53 N·m                        |
| Speed at Pull-out Torque       | 1095 rpm                        |
| Full-load Torque (at 1400 rpm) | 3.13 N·m                        |
| Synchronous Speed              | 1500 rpm                        |
| Rated Speed (nameplate)        | 1400 rpm                        |
| Full-load Slip                 | $(1500 - 1400) / 1500 = 6.67\%$ |

The pull-out torque (5.53 N·m) is 1.77× the rated full-load torque (3.13 N·m), giving a good overload margin. The starting torque (3.32 N·m) is slightly higher than the rated torque, providing adequate starting capability for the rated load.

## 5. Complete Design Parameter Summary

| Design Parameter          | Value                                 |
|---------------------------|---------------------------------------|
| Motor Type                | 3-Phase Squirrel Cage Induction Motor |
| Rated Power               | 0.18 kW                               |
| Supply Voltage            | 415 V (line), 230.94 V (phase, Y)     |
| Rated Current             | 0.73 A                                |
| Rated Speed / Sync. Speed | 1400 rpm / 1500 rpm                   |

|                             |                                                |
|-----------------------------|------------------------------------------------|
| Frequency                   | 50 Hz                                          |
| Number of Poles             | 4                                              |
| Number of Stator Slots      | 24                                             |
| Winding Type                | Single-layer, Fractional Pitch, Distributed    |
| Number of Coils             | 12 (4 per phase)                               |
| Coil Pitch (Mech.)          | 75° (spans 5 slots)                            |
| Pitch Factor $K_p$          | 0.9659                                         |
| Distribution Factor $K_d$   | 0.9659                                         |
| Winding Factor $K_w$        | 0.933                                          |
| Flux per Pole, $\phi$       | $1.48 \times 10^{-3}$ Wb                       |
| Turns per Phase, $N_p$      | 783                                            |
| Turns per Coil, $N_c$       | 196                                            |
| Wire Gauge                  | 29 SWG (0.345 mm dia, 0.0937 mm <sup>2</sup> ) |
| Insulation Class            | F (max temp rise 105°C above 40°C ambient)     |
| NEMA Code Letter            | M                                              |
| Stator Resistance $R_1$     | 32.125 $\Omega$                                |
| Stator Reactance $X_1$      | 17.36 $\Omega$                                 |
| Rotor Reactance $X_2$       | 26.04 $\Omega$                                 |
| Rotor Resistance $R_2$      | 14.325 $\Omega$                                |
| Magnetising Reactance $X_m$ | 160.28 $\Omega$                                |